Lab_Syn_3.0

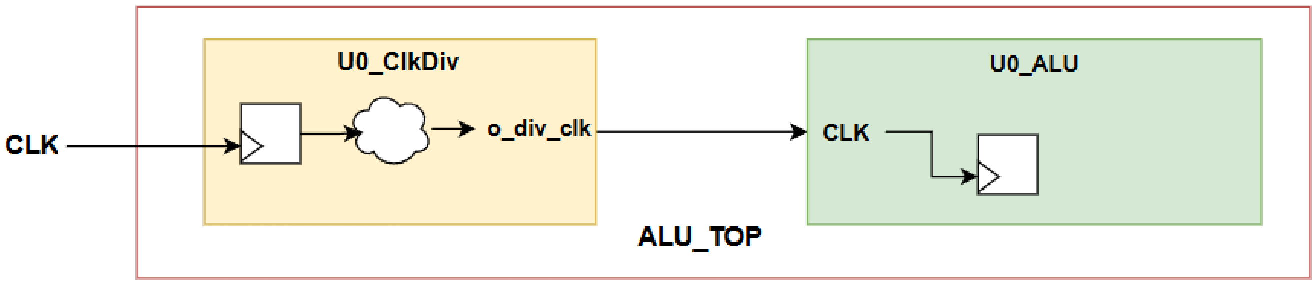
You have to synthesize ALU_TOP hierarchy files using Design Compiler shell and generate Technology Dependent gate level netlists in Verilog format using standard cell libraries (typical, worst, best) inside std_cells folder:-

- scmetro_tsmc_cl013g_rvt_tt_1p2v_25c.db
- scmetro_tsmc_cl013g_rvt_ff_1p32v_m40c
- scmetro_tsmc_cl013g_rvt_ss_1p08v_125c

Steps: -

1. Copy Lab_Syn_3.0 folder from your computer into the virtual machine inside Labs directory
2. Open a terminal inside path /home/ICer/Labs/Lab_Syn_3.0/syn and run syn_script.tcl script through design compiler shell using the following command in the terminal
    - dc_shell -f syn_script.tcl | tee syn.log
3. Check the synthesis log (syn.log) to be sure that: -
    1) No Errors
    2) No Critical Warnings
    3) No Latches
    4) No Comb Loops

- Here is a simple block diagram for **ALU_TOP** which represent master clocks as well generated clocks connections to help you to constraint our top module



4. Open syn_script.tcl file and Add the following constraints in the constraints section before compile command.
    - Create a clock on **CLK** port with frequency 100 MHz
    - Create generated clock on ALU clock with ratio half of master clock frequency
    - Create clock uncertainty with 0.2 ns for setup for all master and generated clocks
    - Create clock uncertainty with 0.1 ns for hold for all master and generated clocks
    - Create a clock transition with 0.05 ns for all master and generated clocks
    - Input delays on all input ports except (CLK & RST) with 20% clock period
    - output delays on all output ports with 20% clock period
    - Add Buffer driving cell for all input ports
    - Add load of 50 pf on all output ports
    - Add dont_touch attribute on CLK & RST ports
    - Set operation condition using slow and fast libraries
    - Set wireload model using slow library

5. Report the following reports and redirect each report to a file
    1) Power report using (report_power -hierarchy) command
    2) Area report using (report_area -hierarchy) command
    3) Setup timing analysis report (report_timing -max_paths 100 -delay_type max) command
    4) Hold timing analysis report (report_timing -max_paths 100 -delay_type min) command
    5) Clocks report using (report_clock –attributes) command
    6) Constraints report using (report_constraint -all_violators) command

6. Generate the following files: -
    1) Gate level netlist in Verilog format
    2) Gate level netlist in ddc format
    3) Standard delay format (SDF) file
    4) Synopsys design constraints (SDC) file

7. Open Design Compiler GUI, View the Design hierarchy.